

# System Requirements Specification

## ReqAgent — AI Requirement Intake & Tracker

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### 1. Introduction

This System Requirements Specification (SRS) defines the complete set of system requirements for ReqAgent. It serves as the definitive reference for development, testing, and acceptance of the system.

| Term                | Definition                                                                 |
|---------------------|----------------------------------------------------------------------------|
| ReqAgent            | The AI-powered requirements intake and tracking system                     |
| LLM                 | Large Language Model — specifically Groq's LLaMA 3.3 70B                   |
| MoSCoW              | Must Have / Should Have / Could Have / Won't Have prioritization framework |
| BA                  | Business Analyst — primary user of the system                              |
| MongoDB             | NoSQL document database used for raw intake storage                        |
| Postgres            | Relational database used for structured output storage                     |
| Traceability Matrix | Document linking requirements to user stories and acceptance criteria      |

### 2. System Overview

ReqAgent is a web-based application built with Python and Streamlit. It accepts business requirement submissions via a structured form, processes them using an LLM API, and stores results in a dual-database architecture. The system provides a 4-tab dashboard for requirement management.

### 3. Functional Requirements

| ID      | Requirement                                              | Priority | BR    |
|---------|----------------------------------------------------------|----------|-------|
| SRS-F01 | System shall accept requirement submissions via web form | High     | BR-01 |

| ID      | Requirement                                                  | Priority | BR    |
|---------|--------------------------------------------------------------|----------|-------|
| SRS-F02 | System shall validate all required fields before processing  | High     | BR-01 |
| SRS-F03 | System shall store raw submission in MongoDB Atlas           | High     | BR-10 |
| SRS-F04 | System shall call Groq API with structured prompt            | High     | BR-02 |
| SRS-F05 | System shall parse AI response as structured JSON            | High     | BR-02 |
| SRS-F06 | System shall generate exactly 3 user stories per submission  | High     | BR-02 |
| SRS-F07 | System shall generate Given/When/Then acceptance criteria    | High     | BR-02 |
| SRS-F08 | System shall assign MoSCoW classification                    | High     | BR-03 |
| SRS-F09 | System shall assign role-weighted priority score             | High     | BR-04 |
| SRS-F10 | System shall detect conflicts against existing requirements  | High     | BR-05 |
| SRS-F11 | System shall store structured output in Neon Postgres        | High     | BR-10 |
| SRS-F12 | System shall support 6-stage lifecycle status management     | High     | BR-06 |
| SRS-F13 | System shall log all status changes with timestamp           | Medium   | BR-07 |
| SRS-F14 | System shall generate filterable traceability matrix         | Medium   | BR-08 |
| SRS-F15 | System shall export traceability matrix as CSV               | Medium   | BR-08 |
| SRS-F16 | System shall display KPI metrics and charts in analytics tab | Medium   | BR-09 |

## 4. Non-Functional Requirements

| ID       | Category        | Requirement          | Acceptance Criteria                  |
|----------|-----------------|----------------------|--------------------------------------|
| SRS-NF01 | Performance     | App page load time   | < 3 seconds                          |
| SRS-NF02 | Performance     | AI processing time   | < 10 seconds per submission          |
| SRS-NF03 | Security        | Credential storage   | Zero credentials in source code      |
| SRS-NF04 | Security        | Database connections | SSL/TLS enforced on all connections  |
| SRS-NF05 | Reliability     | System availability  | 99%+ uptime during business hours    |
| SRS-NF06 | Maintainability | Code modularity      | Separate modules per functional area |
| SRS-NF07 | Portability     | Deployment target    | Deployable to Streamlit Cloud        |
| SRS-NF08 | Scalability     | Concurrent users     | Supports 10+ concurrent users        |

## 5. Interface Requirements

### 5.1 User Interface

- Web-based interface accessible via modern browser
- Dark/light theme with consistent corporate blue color scheme
- Responsive layout supporting desktop and tablet viewports
- 4-tab navigation: Submit, Tracker, Traceability, Analytics
- Form validation with inline error messages

### 5.2 External Interfaces

| Interface     | Type           | Description                                       |
|---------------|----------------|---------------------------------------------------|
| Groq API      | REST API       | LLM processing via llama-3.3-70b-versatile model  |
| MongoDB Atlas | Cloud Database | Raw intake document storage via pymongo driver    |
| Neon Postgres | Cloud Database | Structured relational storage via psycopg2 driver |

## 6. Constraints and Assumptions

### Constraints

- Must use Python 3.12 and Streamlit framework
- Must use Groq API free tier (rate limits apply)
- Must use MongoDB Atlas and Neon Postgres free tiers
- No budget for paid cloud infrastructure

### Assumptions

- Stakeholders submit requirements in English
- Groq API maintains sufficient availability
- Users have access to a modern web browser
- Internet connectivity is available for cloud database access